

WASP-2b

R-0212

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_180804 · created 2026-07-17T00:18:08+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458334.6981 +/- 0.0053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.094 +/- 0.025
Transit depth (Rp/Rs)^2	0.88 +/- 0.47 [%]
Orbital Inclination (inc)	85.03 +/- 1.92
Airmass coefficient 1 (a1)	1.259 +/- 0.013
Airmass coefficient 2 (a2)	-0.2 +/- 0.01
Scatter in the residuals of the lightcurve fit is	1.01 %
Best Comparison Star	None
Optimal Aperture	4.88
Optimal Annulus	17.96
Transit Duration (day)	0.063 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.094 ± 0.025 vs 0.1326 (deviation 1.37σ)
Duration fit vs published	0.063 d vs 0.0737 d (deviation 0.38σ)
Mid-time O–C	-1.8 min
Depth SNR	3.3
Residual scatter	1.01 %

Light curve

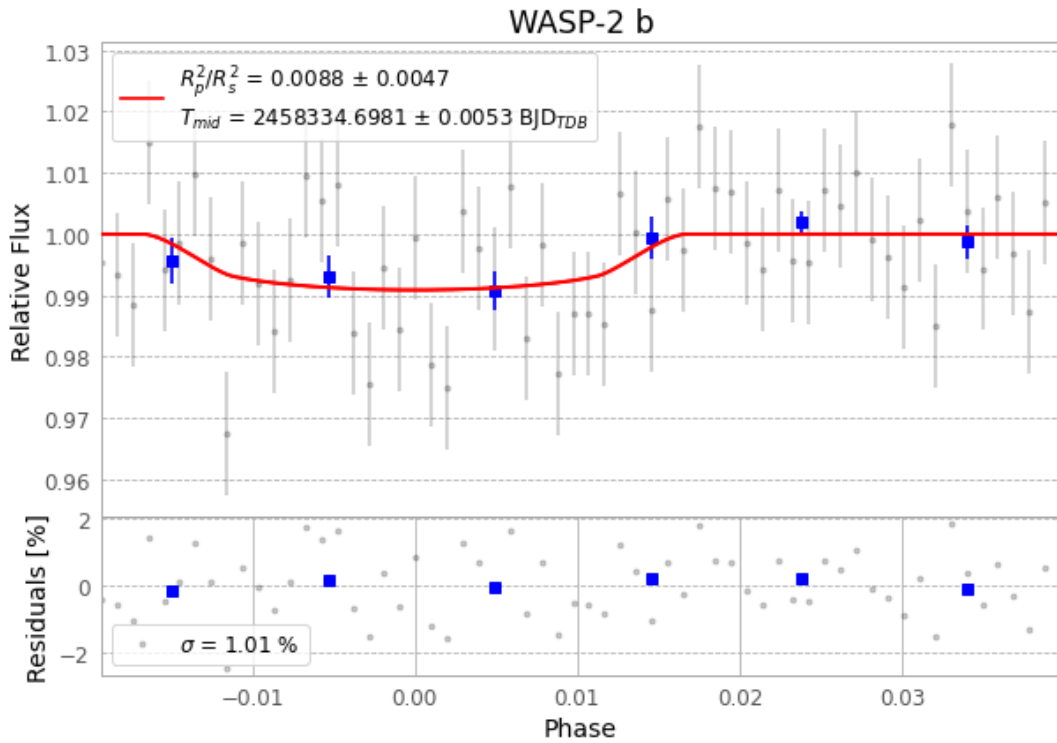


Figure 1 — FinalLightCurve_WASP-2 b_2018-08-03.png (SHA-256 0a3d61e748bd1969...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [309, 166], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a8604cea831108f5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

62 FITS frames (41 MB), WASP-2180804033612.FITS → WASP-2180804063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-180804.log (72b8e9d296b2...), FinalLightCurve_WASP-2 b_2018-08-03.png (0a3d61e748bd...), FinalLightCurve_WASP-2 b_2018-08-03.csv (2674df6c3880...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 00:18Z.